Section 1.1 — Sets and Numbers — Answer Key

Math 103: College Algebra

1 Standard Number Sets

For each number below, circle the symbol for each set the number belongs to.

1. $-\sqrt{7}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	2. $-\frac{75}{5}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	3. $\frac{1}{5}$	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
4. -6	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	5. 5	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	6. 2	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
7. $\frac{7}{3}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	8. $\frac{21}{3}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	9. 3.61	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
10. π	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	11. -4.0	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	12. -4	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
13. $\sqrt{8}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	14. $\sqrt{9}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	15. $-4.\overline{54}$	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
16. -0.2	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	17. -3215	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	18. 9.0	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
19. 816	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	20. $-\frac{10}{7}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	21. $0.\overline{123}$	$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$
22. $-\frac{3}{4}$	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	23. 0.1001110000...	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$		

Note: Answers here will vary. Those given are just some possible examples.

24. -5

25. $\frac{1}{2}$

26. $\sqrt{2}$

2 Set Builder Notation

For each set A given below, list three elements of the set. Then, write the given number x in a form that shows that $x \in A$. *Note: Answers here will vary. Those given are just some possible examples.*

27. Elements: $\frac{1}{8}$, $\frac{2}{8}$, $\frac{3}{8}$. $x = \frac{4}{8} \in A$

28. Elements: $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{6}$. $x = \frac{3}{4} \in A$

29. Elements: $1 = 3 - 2$, $2 = 4 - 2$, $3 = 5 - 2$. $x = 1 - 2 \in A$

30. Elements: $\frac{1}{2}$, $\frac{1}{7}$, $\frac{2}{3} = \frac{1}{\frac{3}{2}}$. $x = \frac{1}{\frac{1}{3}} \in A$

3 Properties of Real Numbers/Simplifying Expressions

Simplify each expression below. Clearly show your work on a separate sheet of paper.

31. $5x$

32. $-6a$

33. $5t - 1$

34. $2x - 8$

35. $2t + 12$

36. $4t + 10$

37. $\frac{1}{2}x + \frac{5}{2}$

38. $-x + 30$

39. $\frac{5}{6}x + \frac{3}{2}$

40. $-3.5x + 8.5$

41. $2.3x - 1.5$

42. $x + y + 14$

43. $25.5x - 73.8$

4 Properties of Equality/Solving Equations

Solve each equation below. Clearly show your work on a separate sheet of paper.

44. $x = 5$
45. $x = -\frac{4}{11}$
46. $y = \frac{20}{3}$
47. $y = 16$
48. No solution
49. $a = 2$
50. All real numbers are solutions
51. $a = -9$
52. No solution
53. $x = -8$
54. All real numbers are solutions
55. $a = -9$
56. $v = 4.1$
57. No solution